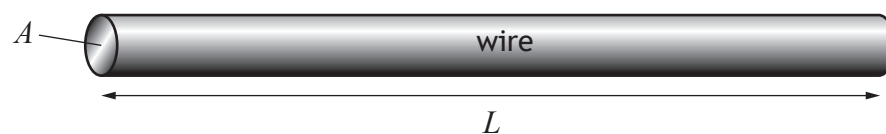


15. A wire of length L and cross-sectional area A is shown.



The resistance R of the wire is given by the relationship

$$R = \frac{\rho L}{A}$$

where ρ is the resistivity of the wire in $\Omega \text{ m}$.

- (a) The resistivity of aluminium is $2.8 \times 10^{-8} \Omega \text{ m}$.

Calculate the resistance of an aluminium wire of length 0.82 m and cross-sectional area $4.0 \times 10^{-6} \text{ m}^2$.

2

Space for working and answer



15. (continued)

MARKS

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MARGIN

- (b) A student carries out an investigation to determine the resistivity of a cylindrical metal wire of cross-sectional area $4.52 \times 10^{-6} \text{ m}^2$.

$$4.52 \times 10^{-6} \text{ m}^2$$



The student varies the length L of the wire and measures the corresponding resistance R of the wire.

The results are shown in the table.

Length of wire L (m)	Resistance of wire R ($\times 10^{-3} \Omega$)
1.5	5.6
2.0	7.5
2.5	9.4
3.0	11.2
3.5	13.2

- (i) Using the square-ruled paper on *Page 36*, draw a graph of R against L . 3
- (ii) Calculate the gradient of your graph. 2
Space for working and answer

- (iii) Determine the resistivity of the metal wire. 3
Space for working and answer

[END OF QUESTION PAPER]



* X 7 5 7 7 6 0 1 3 5 *